Supplement to Fragment-Mixed Continuation for Cross-Corpus Visual Speech Recognition

Fixed greedy CTC readout diagnostic

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1 Question and status

This exploratory diagnostic asks whether the short-only disadvantage persists when the same fitted native full-only (FC), whole-word mixed (WW), and whole-word short-only (SO) states are read using greedy CTC. It was specified after the completed common-decoder short-only analysis. It is not an untouched-test confirmation, a new training seed, or a decoder-selection experiment. The common hybrid beam decoder remains the primary readout in the manuscript.

All three states use seed 17 and the final 6,000-update endpoint. There is no retraining or tuning. The complete populations are 11,307 VLR clips, 225,960 reference units and 61 archived source groups, and 1,167 ViCocktail clips, 30,091 units and 20 source groups. VLR contains 1,634,458 encoded frames and ViCocktail 170,078. Source-group labels are not verified speaker identities.

2 Fixed readout and verification

For every encoded frame, choose the highest CTC log probability, collapse adjacent repeated token IDs, then remove blank ID 0 and end-of-sequence ID 2001. An EOS token does not truncate the path. No blank-logit bias, language model, beam search, attention-head search, or readout sweep is used. Token-to-text mapping, Unicode NFC, lowercasing and whitespace normalization are unchanged. Vietnamese WER uses whitespace-delimited units, which need not be lexical words.

Each model/population runs in a fresh process on the same RTX 3060 workstation, with FP32, TF32 disabled, deterministic algorithms and two intra-op CPU threads. The frozen loader verifies the original complete-clip pixels. Admission used six duration-quantile resource probes per population and a fixed time deadline; probe clips are not extra

assessment observations. Every admitted population is complete, locally backed up and independently recounted before inclusion.

The independent CPU analysis reconstructs every collapsed token sequence and normalized transcript from the saved argmax paths. It recalculates edit counts with a scalar distance matrix and traceback, matches reference, clip, position, frame and pixel identities, and checks terminal hashes and runtime bindings. The saved blank-probability sums are range-checked but cannot be recomputed because full frame logits were not archived. Blank-argmax counts are recomputed exactly from every saved path. These are descriptive readout measurements, not proof of a training mechanism.

3 Results

Table 1: Complete-population WER (percent). Δ is greedy minus common; smaller WER is better.

Corpus	State	Common WER	Greedy WER	Δ (pp)
VLR	FC	38.029	42.477	+4.449
VLR	WW	38.606	43.151	+4.545
VLR	SO	44.844	47.060	+2.216
ViCocktail	FC	67.369	69.187	+1.818
ViCocktail	WW	63.388	64.385	+0.997
ViCocktail	SO	70.715	72.424	+1.708

Table 2: Greedy substitutions, deletions, insertions and total predicted whitespace units.

Corpus	State	S	D	I	Output units
VLR	FC	71889	18524	5569	213005
VLR	WW	68426	23396	5683	208247
VLR	SO	56572	46994	2771	181737
ViCocktail	FC	9660	10595	564	20060
ViCocktail	WW	10163	8896	315	21510
ViCocktail	SO	6734	14960	99	15230

Table 3: Readout descriptions (percent). Output/reference uses transcript units; blank statistics are weighted by all encoded frames, not averaged equally over clips.

Corpus	State	Output/reference	Blank argmax	Mean p(blank)
VLR	FC	94.267	85.530	83.288
VLR	WW	92.161	85.806	83.470
VLR	SO	80.429	87.652	85.241
ViCocktail	FC	66.664	87.293	82.991
ViCocktail	WW	71.483	86.288	81.963
ViCocktail	SO	50.613	90.285	85.312

Table 4: Paired archived-source-group contrasts under the fixed greedy readout. Positive differences favor SO.

Corpus	Contrast	Difference (pp)	95% interval
VLR	FC–SO	-4.583	[-4.874, -4.325]
VLR	WW–SO	-3.909	[-4.172, -3.673]
ViCocktail	FC–SO	-3.237	[-3.826, -2.390]
ViCocktail	WW–SO	-8.039	[-9.668, -6.163]

On VLR, SO minus FC is +4.583 WER points and SO minus WW is +3.909 points. The corresponding deletion-count differences are +28470 and +23598, respectively.

On ViCocktail, SO minus FC is +3.237 WER points and SO minus WW is +8.039 points. The corresponding deletion-count differences are +4365 and +6064, respectively.

4 Uncertainty and interpretation

The two planned contrasts within each population are FC minus SO and WW minus SO WER. Positive values favor SO. Intervals use 10,000 paired resamples of the archived source groups, NumPy's default generator with seed 20261031, and linear percentile endpoints at 2.5 and 97.5. Resampling preserves the shared clips and denominator within each drawn group. Sixteen draws per contrast are checked again by scalar summation. Intervals are pointwise, unadjusted for multiplicity, and conditional on the fitted states. They do not describe training-seed uncertainty or establish causal mechanisms.

Greedy-minus-common differences compare the same state on the same population; they do not authorize replacing the main decoder with whichever scores better. The diagnostic retains the reused-test-set, automatic-timing, single-family, single-control-seed and unequal frame/token-exposure limitations of the main study. A blank-heavy or shorter prediction is an observed property of this readout, not by itself an explanation of how continuation changed the model.

5 Reproduction

The accompanying package includes the frozen diagnostic protocol and inference driver, original cohort and pixel identities, every saved path and transcript, all terminal receipts, per-clip counts, full-precision aggregates and intervals, and the CPU recount program. See its README for the portable commands. Neural inference additionally requires the main research-code package and the exact licensed data/checkpoints; the CPU recount does not rerun the recognizer.